

Surrogate model of DEM simulation for binary-sized particle mixing and segregation

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Fig. S1 shows the temporal change in the particle kinetic energy over time under randomly mixed initial condition computed by the DEM. As a result, the particle kinetic energy reached quasi-steady state after 2.0 second.

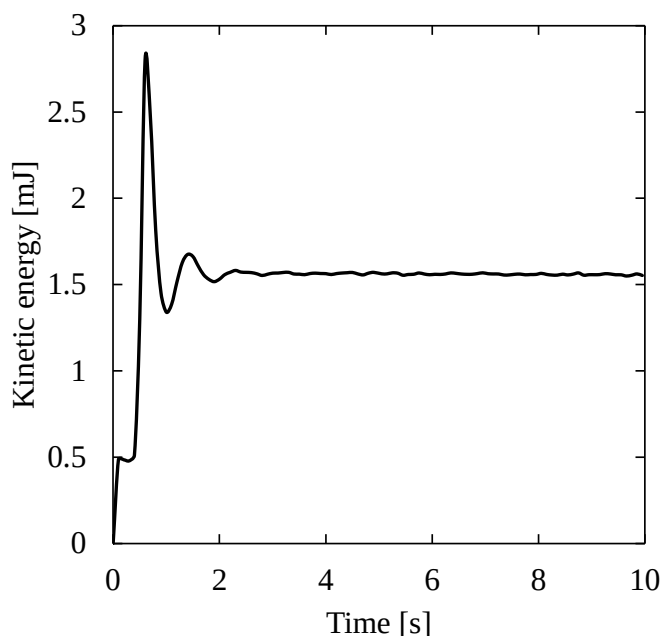


Fig. S1. Temporal change in the total kinetic energy of particles computed by the DEM. The initial condition was randomly mixed initial condition.

Fig. S2 shows the evolution of the training and test loss values over epochs at the Step 3 in case 1. As a result, the training and test loss values were sufficiently small value in the training period. This result indicated that the RNNSR was well trained to predict the local mean component of particle trajectory with high accuracy even in binary-size particle system.

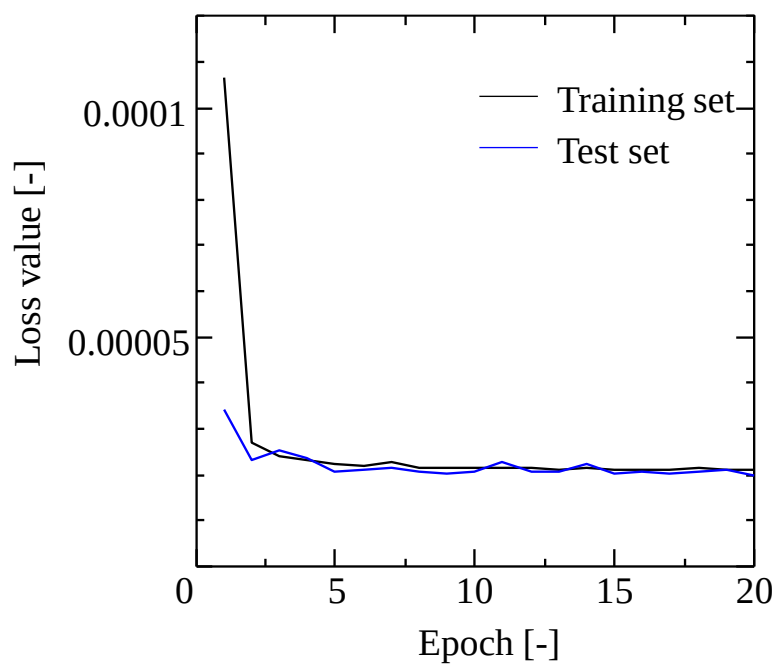


Fig. S2. Evolution of training and test loss of the RNNSR in the case 1.

Fig. S3 shows the Lacey's mixing index at 10 s obtained from the DEM simulation under randomly mixed initial condition. The cell length was varied from 0.025 to 0.1 times the drum diameter (D). The results showed that the Lacey's mixing index significantly increased (0.65 to 0.85) when cell length was higher than $0.04D$, whereas the increase in the mixing index was almost negligible (0.6 to 0.65) for cell lengths below $0.04D$.

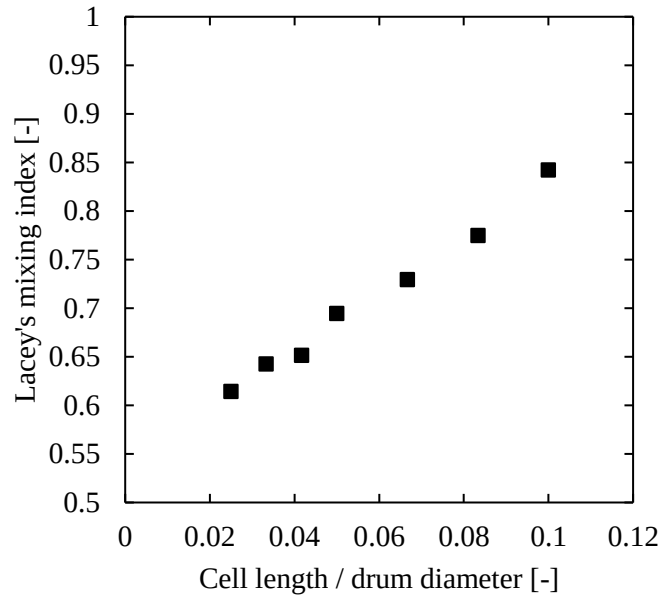


Fig. S3. Sensitive analysis of cell lengths to the Lacey's mixing index at 10.0 s under randomly mixed initial condition. The cell length was varied from 0.025 to 0.1 times the drum diameter (D).